

Opportunistic Communication in Wireless Networks

David Tse
Department of EECS, U.C. Berkeley

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Communication over Wireless Channels

- Fundamental characteristic of wireless channels: fading.
- A modern view of communication over fading channels is emerging.
- This view has ramifications to the design of not only the physical layer but to the design of the entire wireless network.

Opportunistic Communication

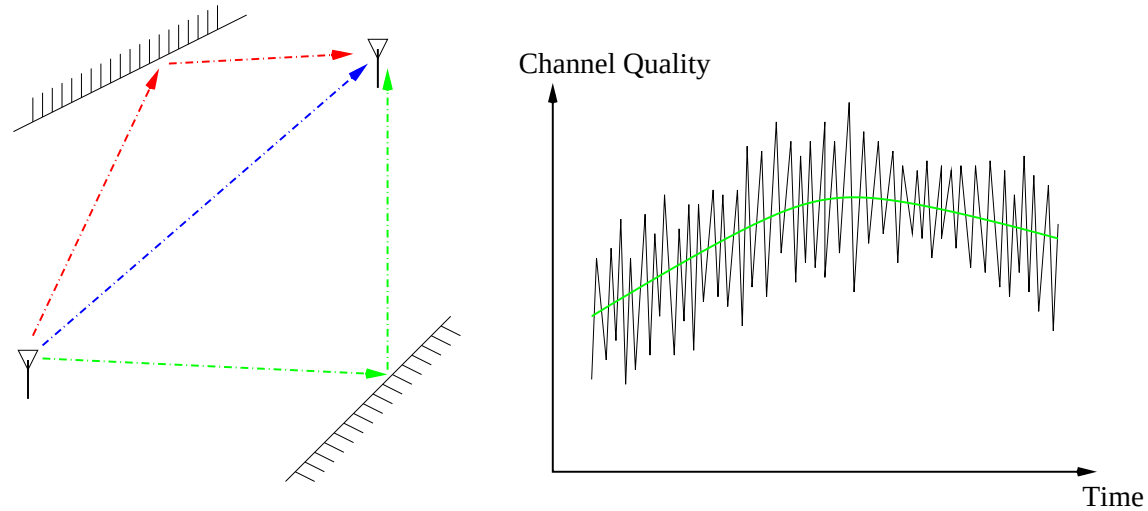
Opportunistic Communication

- Downlink scheduling for Qualcomm's HDR (High Data Rate) system. (Tse 99)

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- Opportunistic beamforming using dumb antennas (Viswanath, Tse and Laroia 2001)

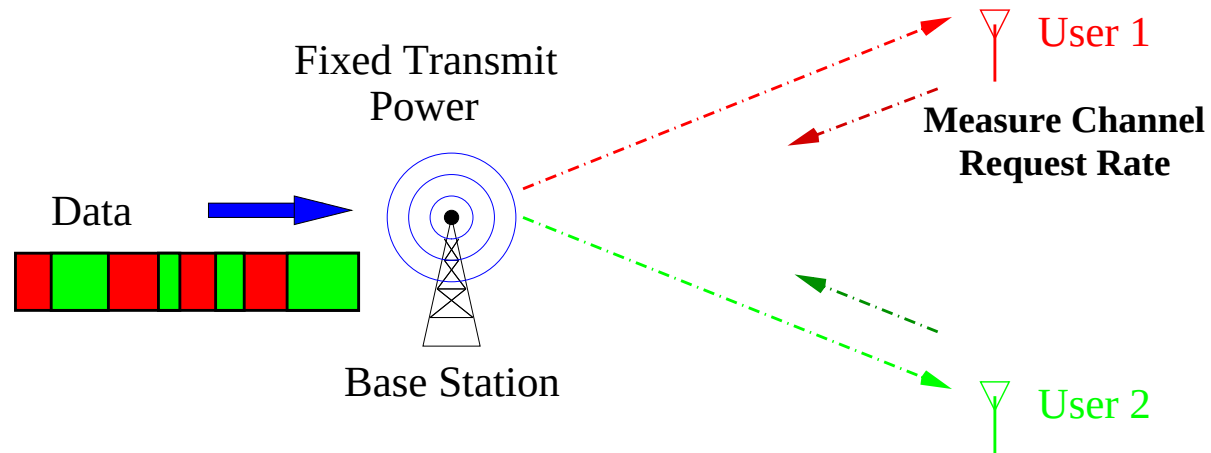
Wireless Fading Channels



- fading due to constructive and destructive interference between multiple signal paths;
- **Rayleigh** fading: superposition of many small paths
- **Rician** fading: many small paths plus one dominant path

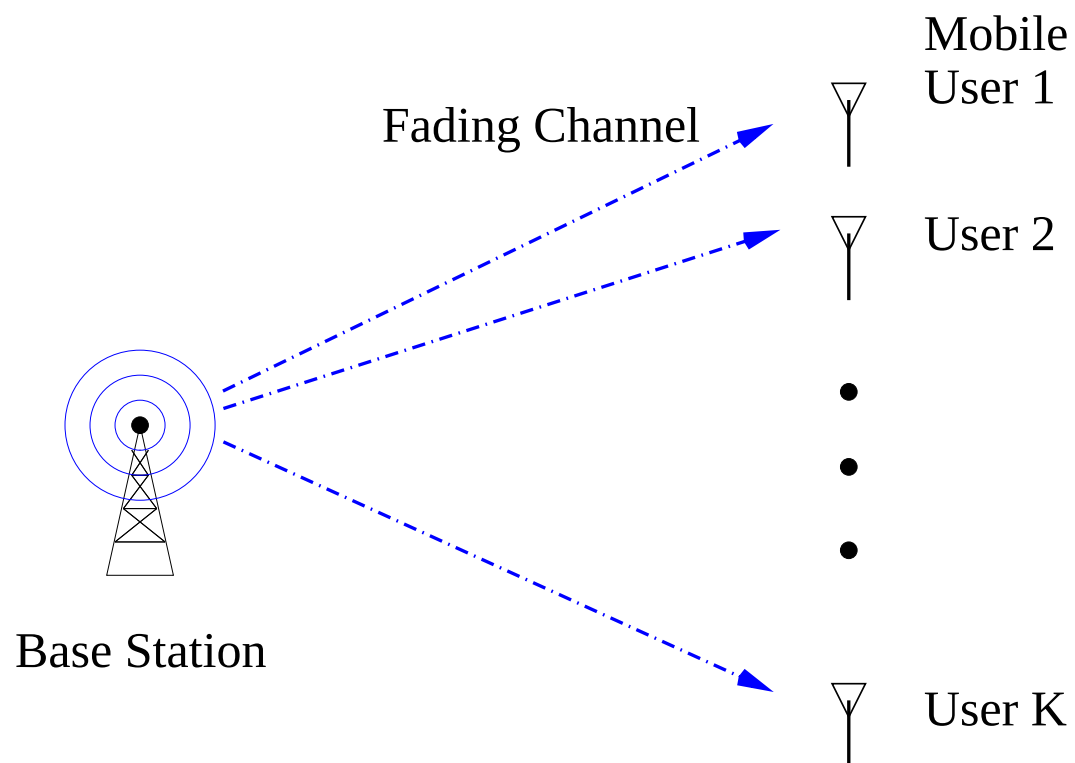
Qualcomm HDR's DownLink

HDR (1xEV-DO): a wireless data system operating on IS-95 band (1.25 MHz)



- HDR downlink operates on a time-division basis.
- Scheduler decides which user to serve in each time-slot.

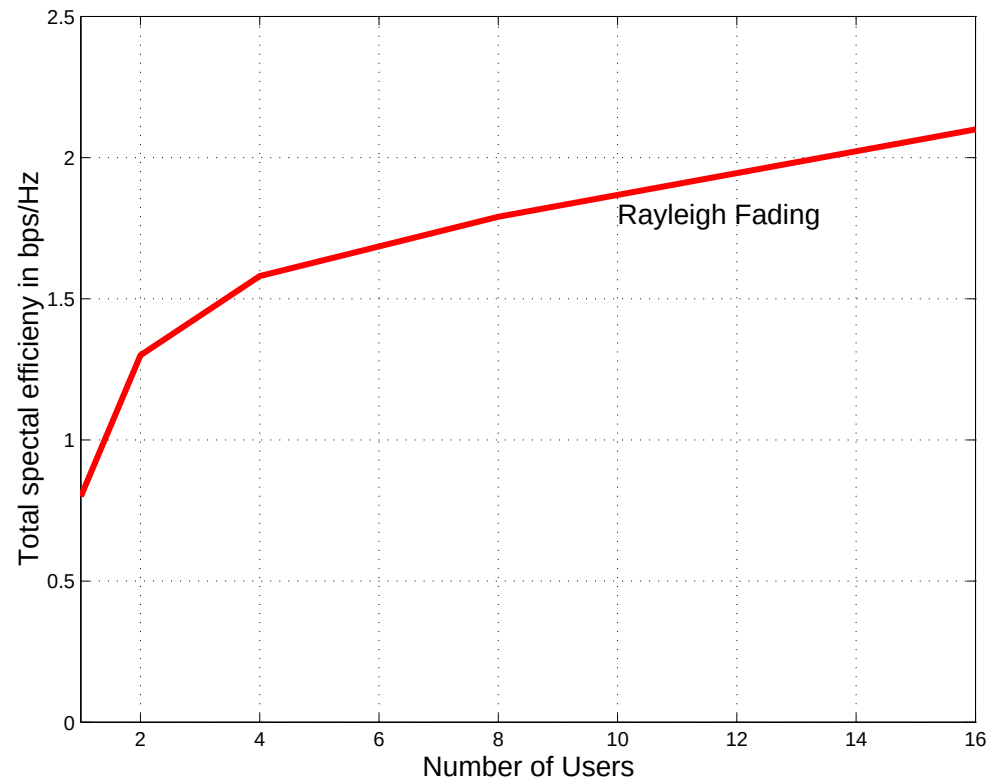
Downlink Multiuser Fading Channel



What is the sum capacity with channel state feedback?

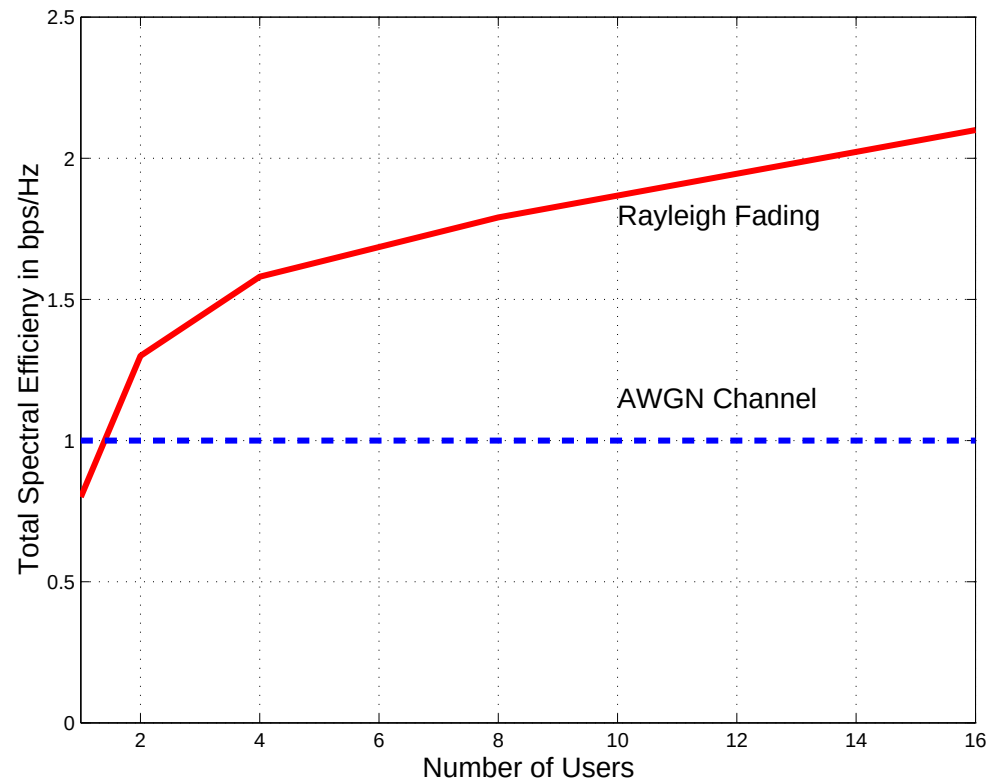
Information Theoretic Capacity of Downlink

(Tse 97)



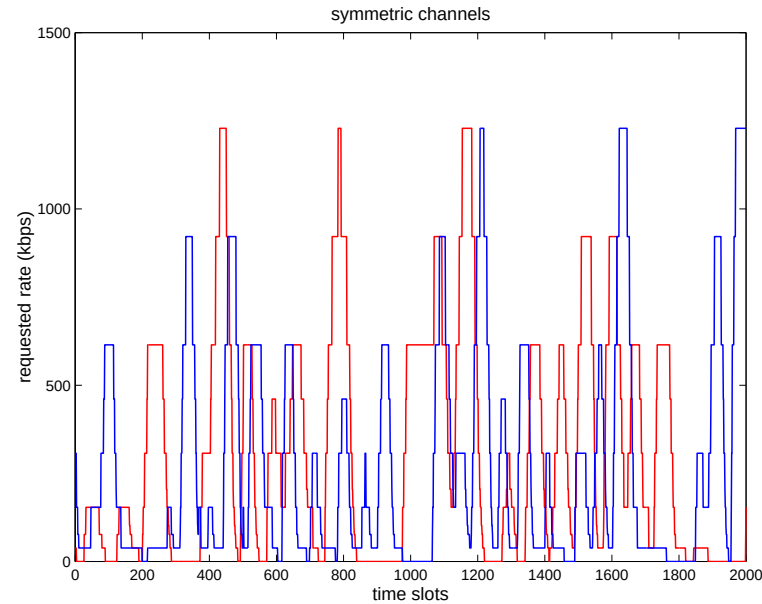
Each user undergoes independent Rayleigh fading with average received signal-to-noise ratio $\text{SNR} = 0\text{dB}$.

To Fade or Not to Fade?



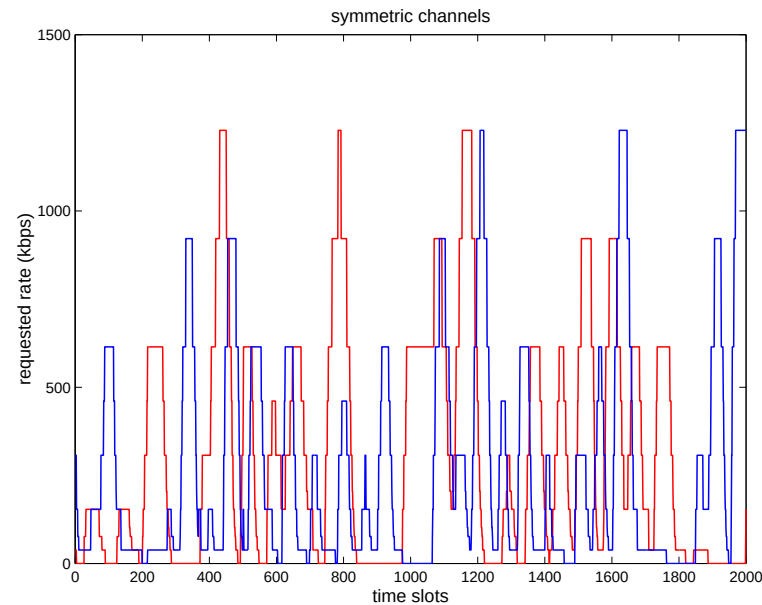
Sum Capacity of fading channel much larger than non-faded channel!

Multiuser Diversity



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- Long term total throughput can be maximized by always serving the user with the **strongest** channel.

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$$\text{effective SNR at time } t = \max_{1 \leq k \leq K} |h_k(t)|^2.$$

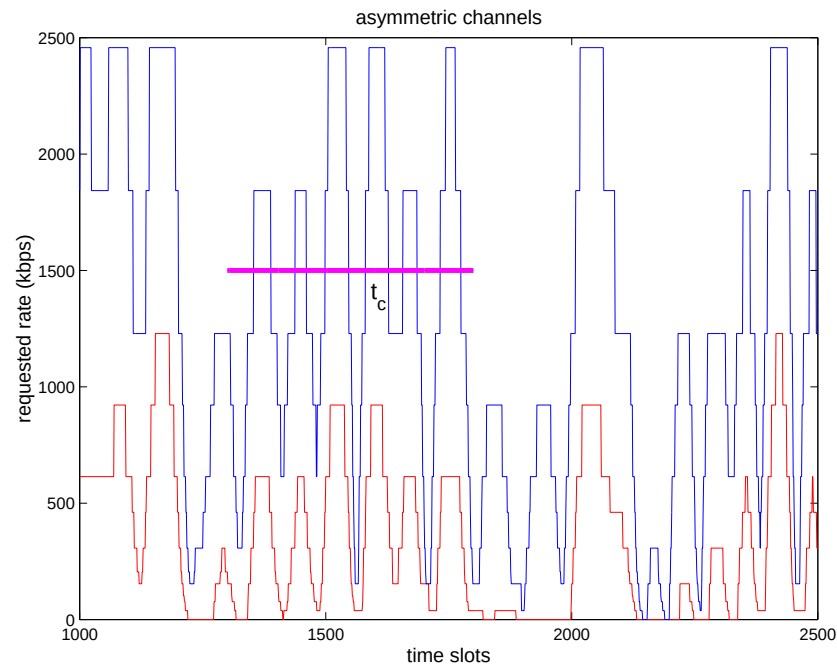
Multiuser Diversity

- Diversity in wireless systems arises from independent signal paths.
- Traditional forms of diversity includes time, frequency and antennas.
- Multiuser diversity arises from independent fading channels across different users.

Multiuser Diversity

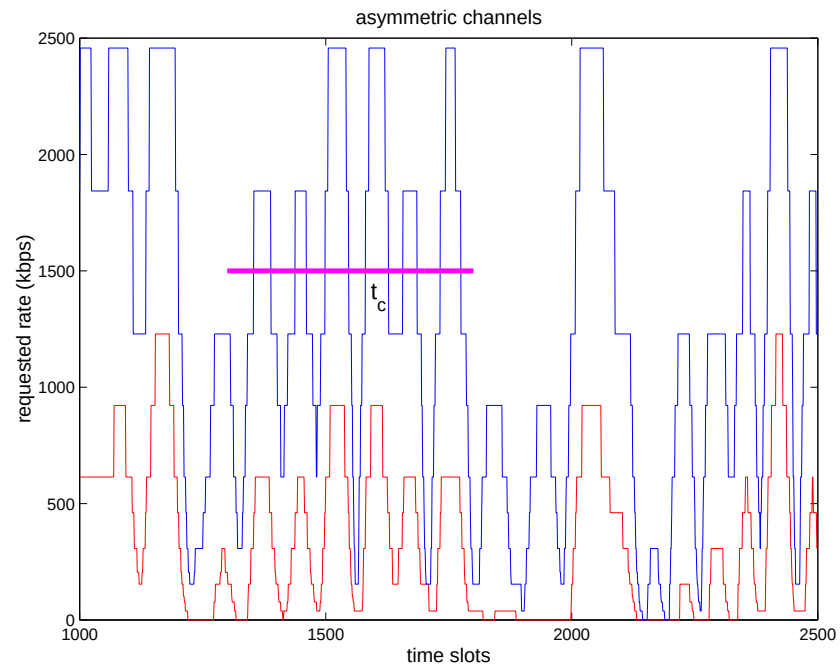
- **Diversity** in wireless systems arises from independent signal paths.
- Traditional forms of diversity includes time, frequency and antennas.
- Multiuser diversity arises from independent fading channels across different users.
- **Fundamental difference**: Traditional diversity modes pertain to **point-to-point** links, while multiuser diversity provides **network-wide** benefit.

Fairness and Delay



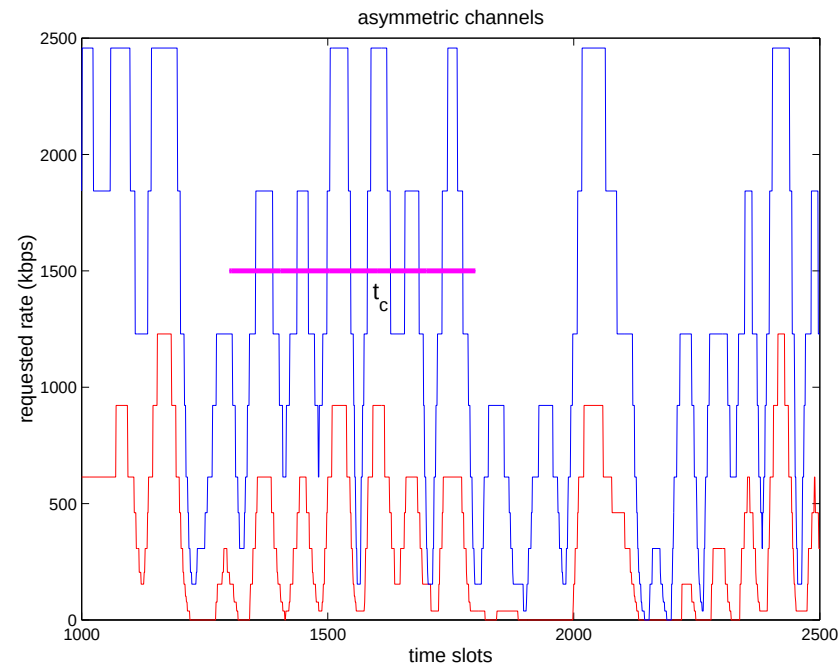
Challenge is to exploit multiuser diversity while sharing the benefits **fairly** and **timely** to users with **asymmetric** channel statistics.

Hitting the Peaks



- Want to serve each user when it is near its **peak** within a latency time-scale t_c .

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- Want to serve each user when it is near its **peak** within a latency time-scale t_c .
- In a **large** system, at any time there is likely to be a user whose channel is near its peak.

Proportional Fair Scheduler

At time slot t , given

- 1) users' average throughputs $T_1(t), T_2(t), \dots, T_K(t)$ in a past window.
- 2) current requested rates $R_1(t), R_2(t), \dots, R_K(t)$

transmit to the user k^* with the largest

$$\frac{R_k(t)}{T_k(t)}.$$

Average throughputs $T_k(t)$ can be updated by an exponential filter with time constant t_c .

Comments

- If users have symmetric channel statistics, this reduces to the greedy policy of transmitting to the mobile with the highest requested rate.
- If channels have different statistics, competition for resource is made fair by normalization
- feedback is built into the metric $R_k(t)/T_k(t)$ to provide a fair bandwidth allocation over the time-scale t_c .

Comparison with Round-Robin Policy

Round-Robin Policy

- Give same number of time slots to all the users in a round-robin fashion, regardless of their channel conditions.

Proportional fair policy:

- Give roughly the same number of time slots to all users, but try to transmit to a user when its channel condition is near its peak.
- Resource fair, but not necessarily performance fair.

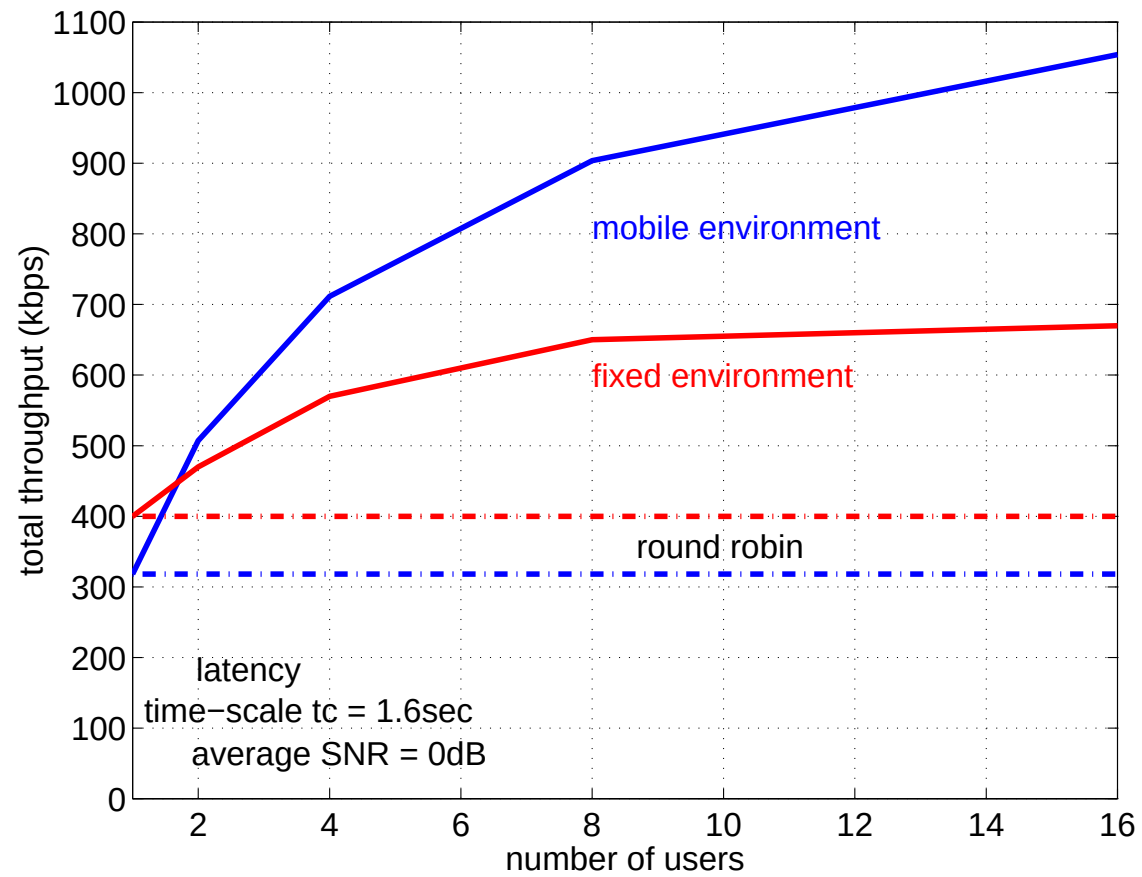
Proportional Fairness

Under stationary assumptions, long-term average throughputs $T_1^*, \dots, T_K^*$ of the scheduler maximizes

$$\sum_k \log T_k$$

among all schedulers.

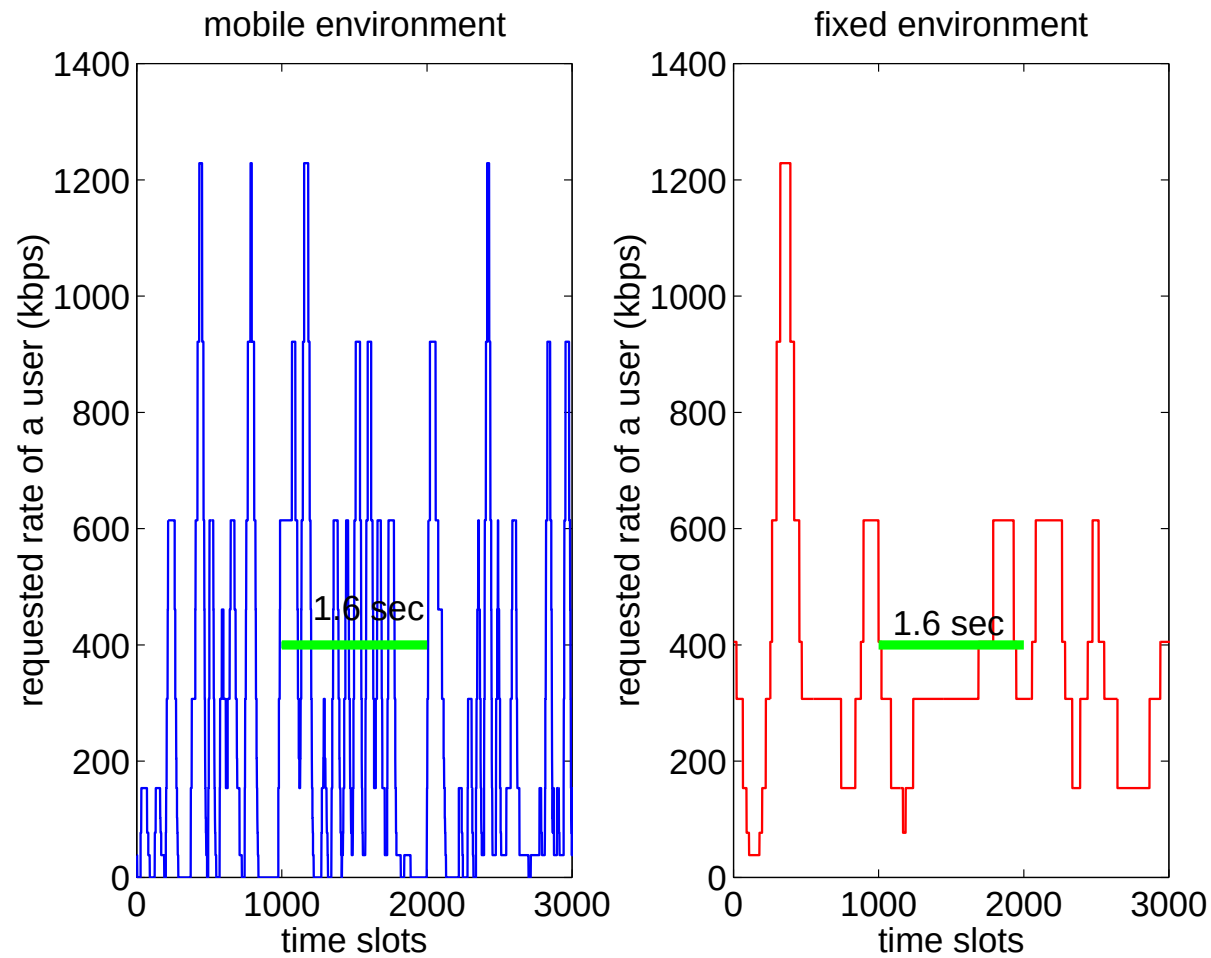
Throughput of HDR Scheduler: Symmetric Users



Mobile environment: 3 km/hr, Rayleigh fading

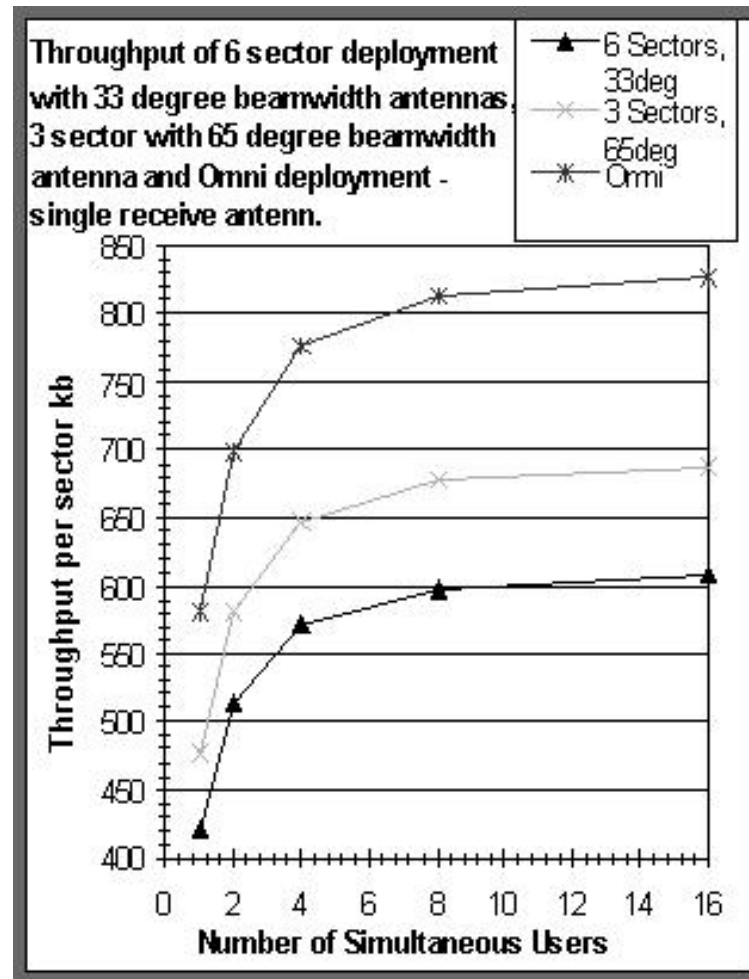
Fixed environment: 2Hz Rician fading with $E_{\text{fixed}}/E_{\text{scattered}} = 5$.

Channel Dynamics



Channel varies faster and has more dynamic range in mobile environments.

Throughput of Scheduler: Asymmetric Users

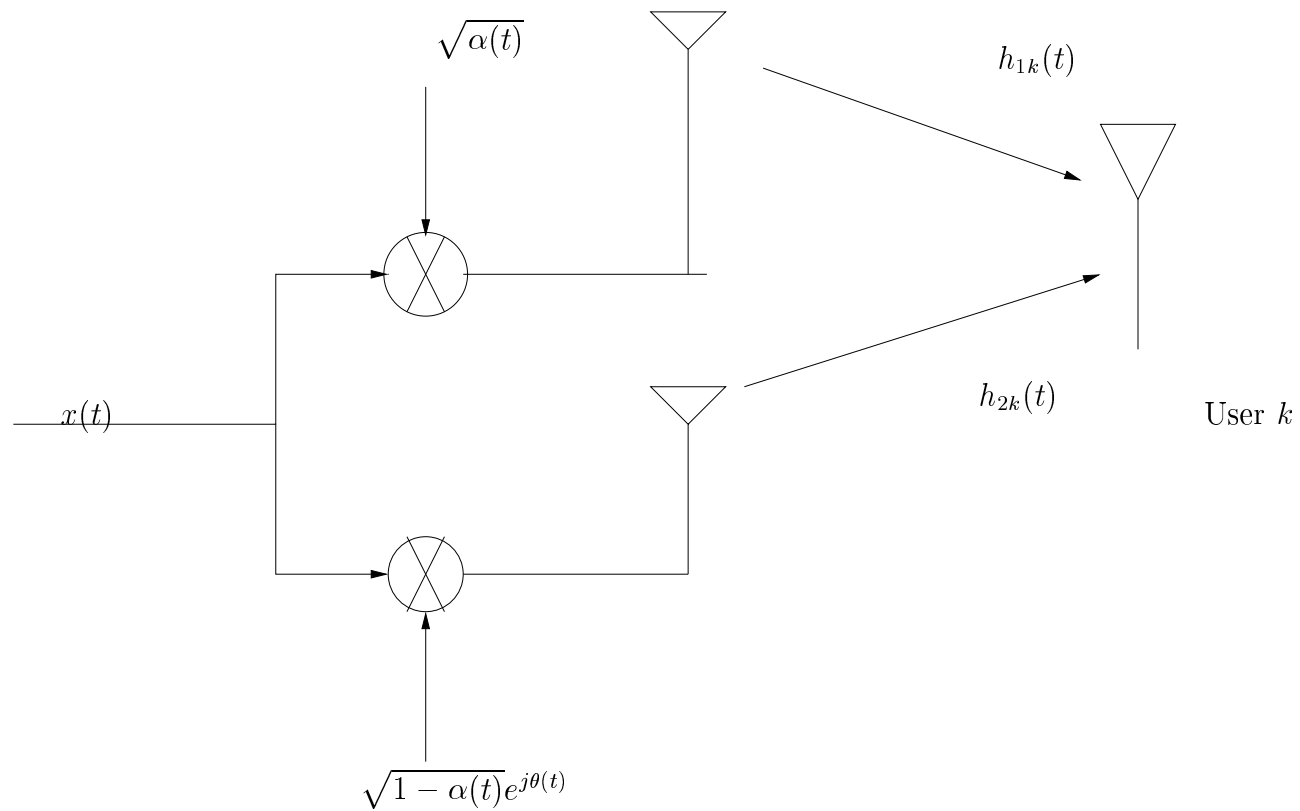


(Jalali, Padovani and Pankaj 2000)

Inducing Randomness

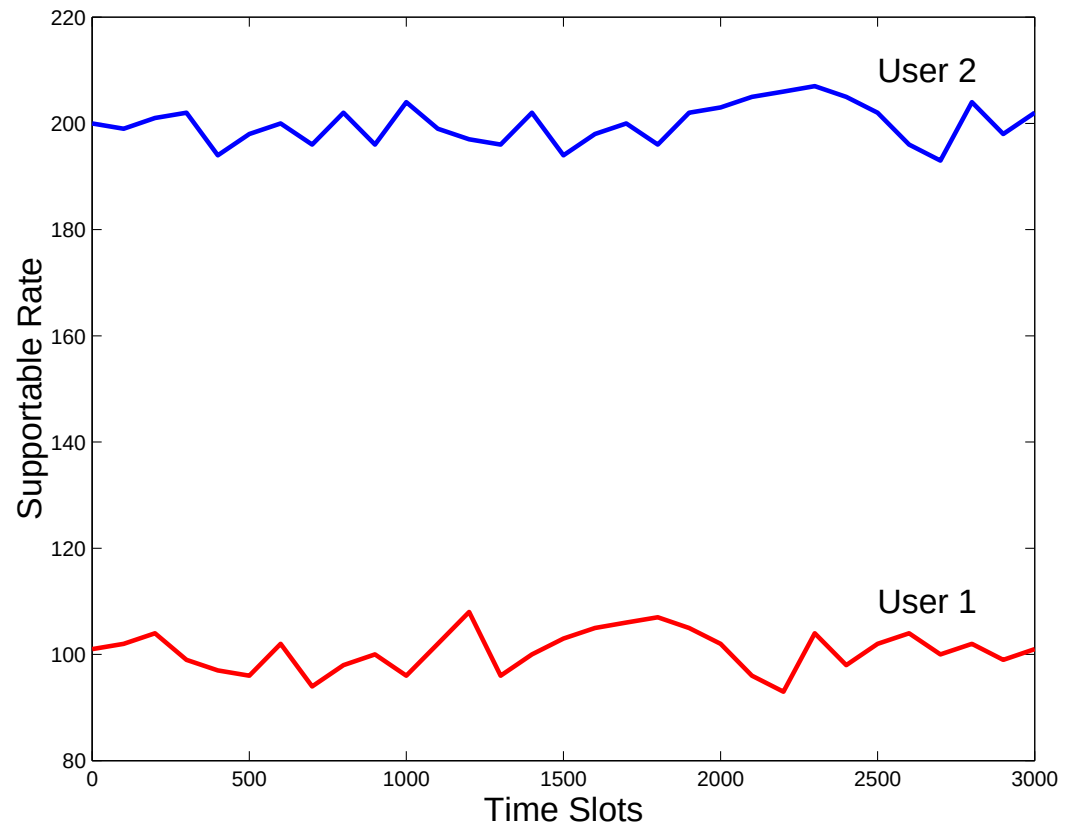
- Scheduling algorithm exploits the nature-given channel fluctuations by **hitting the peaks**.
- If there are not enough fluctuations, why not purposely **induce** them?

Dumb Antennas

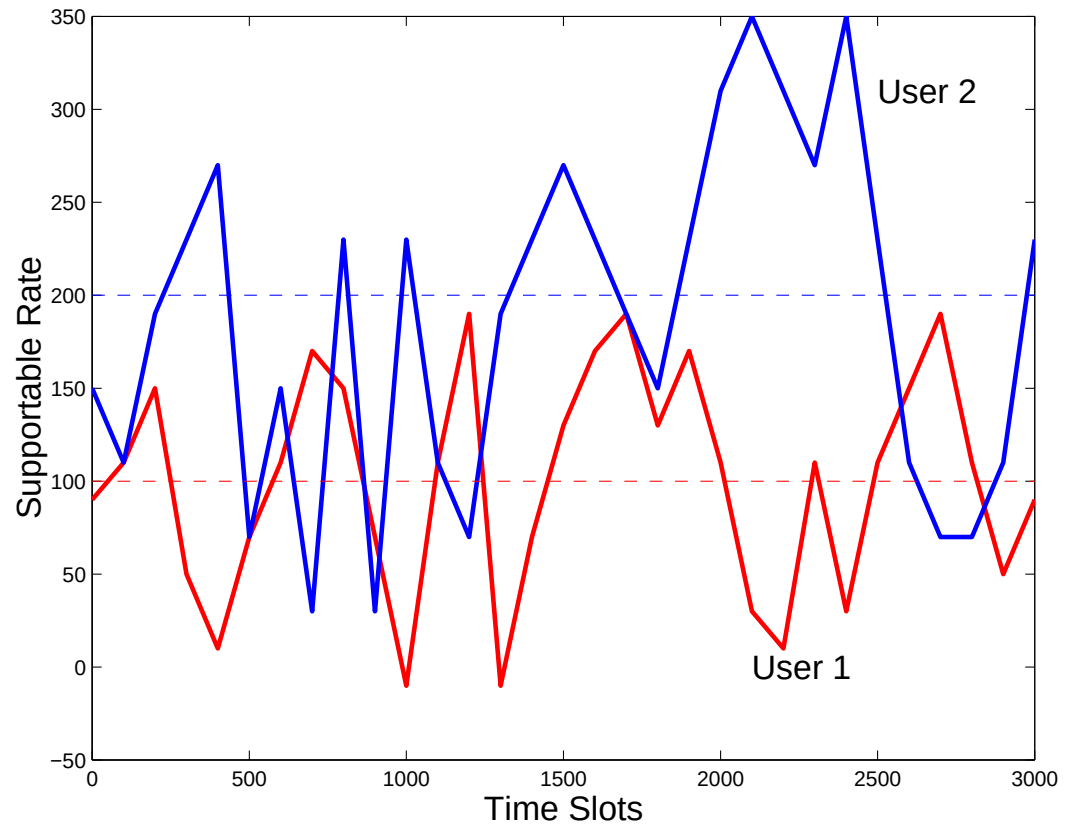


Received signal at user k : $\left[\sqrt{\alpha(t)}h_{1k}(t) + \sqrt{1 - \alpha(t)}\exp(j\theta(t))h_{2k}(t) \right] x(t).$

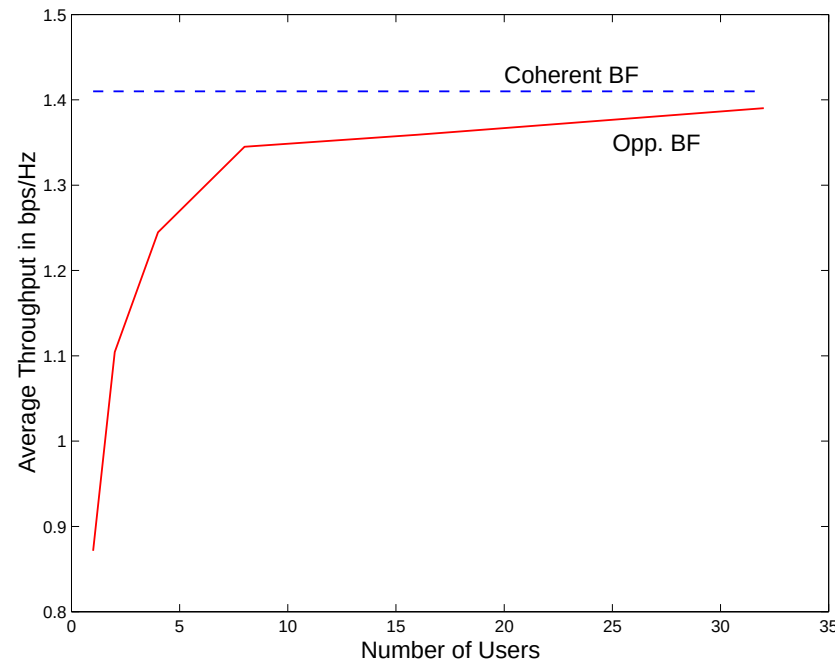
Slow Fading Environment: Before



After



Opportunistic Beamforming: Slow Fading



- Consider first a slow fading environment when channels of the users are fixed (but random).
- Dumb antennas can approach the performance of **true** beamforming when there are many users in the systems.

Opportunistic versus True Beamforming

- If the gains h_{1k} and h_{2k} are known at the transmitter, then **true beamforming** can be performed:

$$\alpha = \frac{|h_{1k}|^2}{|h_{1k}|^2 + |h_{2k}|^2}$$
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- Dumb antennas randomly sweep out a beam and opportunistically sends data to the user closest to the beam.
- Opportunistic beamforming can approach the performance of true beamforming when there are many users in the systems, but with much less feedback and channel measurements.

Asymptotic Result

Assume that the slow fading states of each user are i.i.d. randomly generated (but fixed for all time).

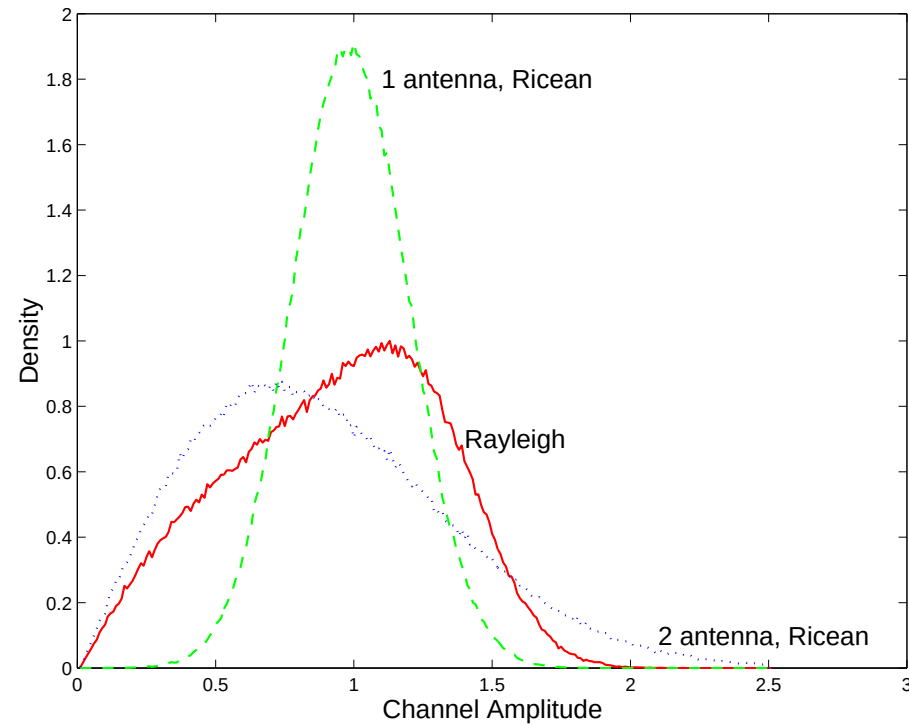
In a large system of K users, with high probability, the users achieve throughputs

$$T_k \rightarrow \frac{1}{K} R_k^{\text{bf}}, \quad k = 1, \dots, K$$

.

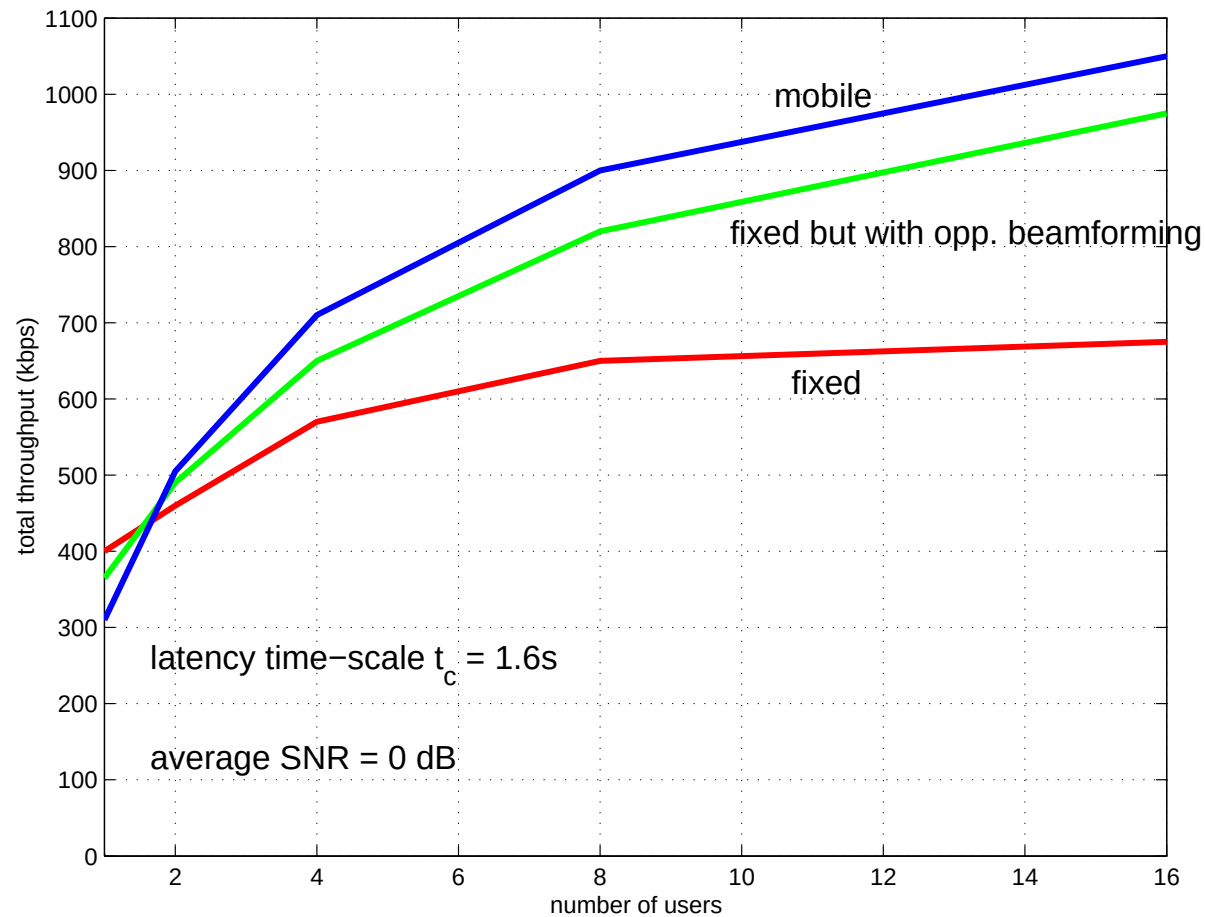
where R_k^{bf} is the rate user k gets when it is perfectly beamformed to.

Opportunistic Beamforming: Fast Fading



Improves performance in fast fading Rician environments by spreading the fading distribution.

Overall Performance Improvement



Mobile environment: 3 km/hr, Rayleigh fading

Fixed environment: 2Hz Rician fading with $E_{\text{fixed}}/E_{\text{scattered}} = 5$.

Comparison to Space Time Codes

- Space time codes: intelligent use of transmit diversity to improve reliability of point-to-point links.
- In contrast, opportunistic beamforming requires no special multi-antenna encoder or decoder nor MIMO channel estimation.
- In fact the mobiles are completely oblivious to the existence of multiple transmit antennas.
- Antennas are truly **dumb**, but yet can surpass performance of space time codes.

Cellular System: Opportunistic Nulling

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- Multiuser diversity allows **interference avoidance**.
- Dumb antennas provides **opportunistic nulling** for users in other cells.
- Particularly important in interference-limited systems with **no** soft handoff.

Traditional CDMA Downlink Design

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- Important for voice with very tight latency requirements.

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- Wants **large** and **fast** fluctuations of both channel and interference so that we can **ride the peaks**.
- Exploits more relaxed latency requirements of **data** as well as MAC layer packet scheduling mechanisms.

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- Rely on sophisticated physical layer signal processing techniques: smart antennas, interference suppression, etc.....
- Future progress will come from taking a broader **network** perspective.

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- Role of MIMO systems (multiple transmit and receive antennas) in multiuser multi-cell wireless networks.

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- Fast channel state feedback with minimal overhead.
- Multiuser diversity and opportunistic nulling as **interference avoidance** techniques.
- Role of MIMO systems (multiple transmit and receive antennas) in multiuser multi-cell wireless networks.
- Fundamental tradeoff between diversity, spatial multiplexing and interference suppression capability.